From Quantum Decoherence to Social Polarisation

A Thermodynamic Model of Public Opinion
and a Proposed Metrological Index

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Abstract

The larger a quantum system, the faster it decoheres. The larger a population, the harder it is to reach agreement. This note formalises the structural analogy between these two phenomena and tests its consequences numerically.

With the Helmholtz free energy $F = E - TS$ written out in full, the stationary opinion distribution exhibits a *phase transition*: above a critical social temperature, a single peak centred on moderation; below it, two extreme peaks. Below that temperature, removing the media field does not restore neutrality — group conformity sustains a *hysteresis*, which explains the ineffectiveness of passive retraction. Finally, when algorithmic gain exceeds a piece of content's natural damping rate ($\gamma\alpha > \lambda$), effective damping becomes negative and the system enters resonance — an informational Larsen effect.

We derive two instruments: the **Entropic Dissipation Index** (EDI), an auditable metric of a feed's informational diversity, and the **Entropic Dissipation Algorithm** (EDA), a recommender filter that optimises this index rather than raw engagement.

Section 8 states the work's limitations, chief among them that the parameters have no empirical calibration procedure. All code, tests and figures are reproducible in containers.

1 Introduction and hypothesis

Sociophysics has applied the tools of statistical physics to opinion dynamics since the 1980s [Galam, 2004, Castellano et al., 2009]. This note starts from a narrower and more precise observation.

In quantum mechanics, an isolated system exists in a superposition of states. On contact with an environment it becomes entangled with it and the interference terms vanish: this is decoherence [Zeh, 1970, Zurek, 2003]. The more massive the environment, the faster the process.

The working hypothesis is:

> Increasing the size of a system ($N \to \infty$) destroys the coherence of its initial state — in quantum physics by accelerating decoherence, in opinion dynamics by making organic consensus unreachable.

The hypothesis is *structural*: it claims that one formalism describes both situations, not that human opinion obeys physical law. Its value lies in yielding measurable quantities. As we shall see (§8), it nonetheless requires an important reformulation: the effect of N is not to add disorder, but to remove plasticity.

2 Entropic formalism

2.1 Two entropies, one quantity

The von Neumann entropy [von Neumann, 1932]

$$S(\rho) = -\operatorname{Tr}(\rho \ln \rho) \tag{1}$$

vanishes for a pure state. It is computed by diagonalisation: the eigenvalues of ρ form a classical distribution to which the Shannon entropy [Shannon, 1948] applies,

$$H(X) = -\sum_i p_i \log_2 p_i. \tag{2}$$

This is precisely what licenses the bridge between the two disciplines: von Neumann entropy *is* the Shannon entropy of the mixture revealed in the eigenbasis.

2.2 A necessary clarification

One common and incorrect formulation must be set aside at once. The evolution of a *closed* quantum system is unitary, so its von Neumann entropy is constant. What grows under decoherence is the entropy of the *reduced subsystem*, obtained by tracing over environmental degrees of freedom.

A clarification follows that strengthens the analogy rather than weakening it: a coherent superposition is a *pure* state of zero entropy. It is therefore not the multiplicity of possibilities that produces disorder, but the loss of coherence between them. The social counterpart is more accurate this way: a population where everyone keeps several opinions open is not disordered — disorder arises from contact with an environment that fixes positions.

3 Thermodynamic dynamics

3.1 The Ising model and social temperature

Each individual carries a binary opinion $s_i \in \{-1, +1\}$ and the population follows the Gibbs-Boltzmann distribution [Ising, 1925]:

$$P(\sigma) = \frac{1}{Z} \exp\left(\frac{J \sum_{\langle i,j \rangle} s_i s_j + H \sum_i s_i}{k_B T} \right), \tag{3}$$

where J measures local conformity, T is the *social temperature* (agitation, irrationality, openness to fluctuation) and H the external media field.

The model's value lies not in the analogy but in a verifiable property: the existence of a critical temperature. In two dimensions its exact value is known [Onsager, 1944]:

$$\frac{T_c}{J} = \frac{2}{\ln(1+\sqrt{2})} \approx 2.269. \tag{4}$$

This is the only point in this work where an exact theoretical prediction, independent of our sociological assumptions, can validate the implementation. Figure 1 recovers it numerically; the residual gap is a finite-size effect on a 24×24 lattice.

3.2 Free energy

The system minimises its Helmholtz free energy $F = E - TS$, where E represents the tension of disagreement and TS the dispersive force. As N grows, configurational entropy grows with it and the TS term eventually dominates E.

3.3 Social hysteresis

This is the most directly actionable result. Below T_c, switching off the media field ($H \to 0$) does not return magnetisation to zero: the interaction term $-J \sum s_i s_j$ takes over and the group sustains the belief through internal cohesion.

The cycle reads in three stages, which are those of a media crisis: injection of a massive field, official retraction, then the need for a counter-field $-H$ exceeding the group's coercive field. Figure 2 shows the cycle area decreasing with social temperature and vanishing above T_c.

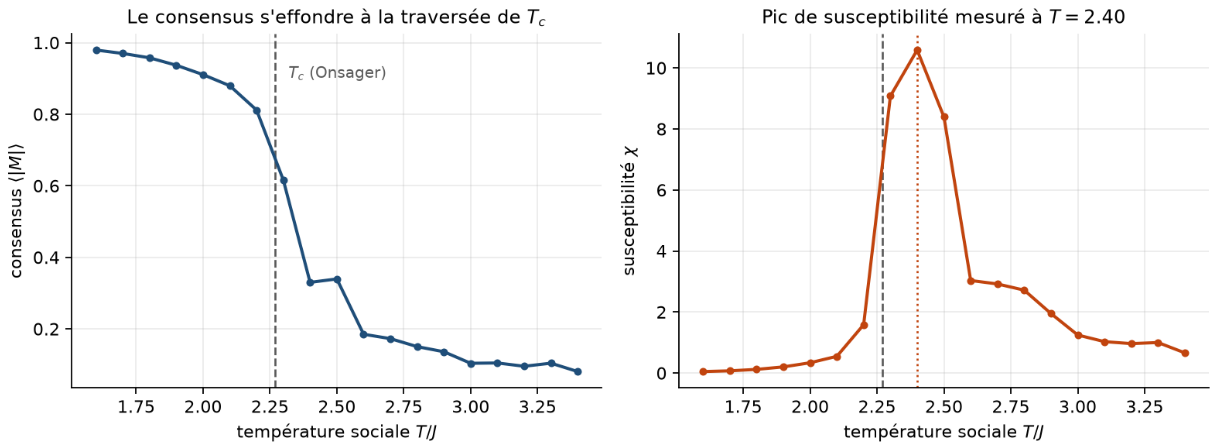

Figure 1: Consensus and susceptibility as functions of social temperature (Monte Carlo, 24×24 lattice). The susceptibility peak locates the transition. Its upward shift relative to Onsager's value (4) is a finite-size effect, not an implementation error. Figure captions are in English; notebook prose is in French.

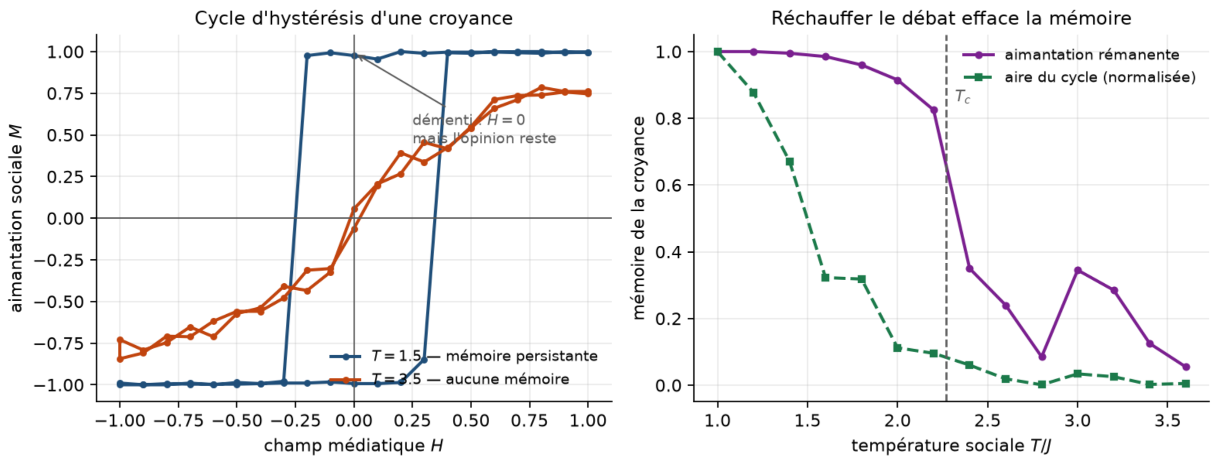

Figure 2: Left: hysteresis cycles below and above T_c; in the former, opinion remains magnetised while the field is zero. Right: decay of belief memory with social temperature — the basis for the thermal-noise recommendation.

Practical consequence: passive debunking does not work. Restoring informational neutrality leaves the belief in place; only active counter-information, or a rise in social temperature, dissipates it.

4 Size effects: consensus time

The voter model [Clifford and Sudbury, 1973, Holley and Liggett, 1975] isolates pure social contagion, without temperature: at each interaction an individual adopts a randomly chosen neighbour's opinion. It yields exact scaling laws for consensus time, measured in sweeps (N elementary interactions).

Measurements (figure 3) give an exponent ≈ 1 in mean field and ≈ 2 on a ring. **A densely connected network therefore converges faster than a geographic neighbourhood.**

This invalidates a frequent argument, that the small-world structure of social networks [Watts and Strogatz, 1998] prevents consensus. Connectivity does not fragment: it accelerates and amplifies, in whatever direction the field gives it. The causes of fragmentation are the *directional*

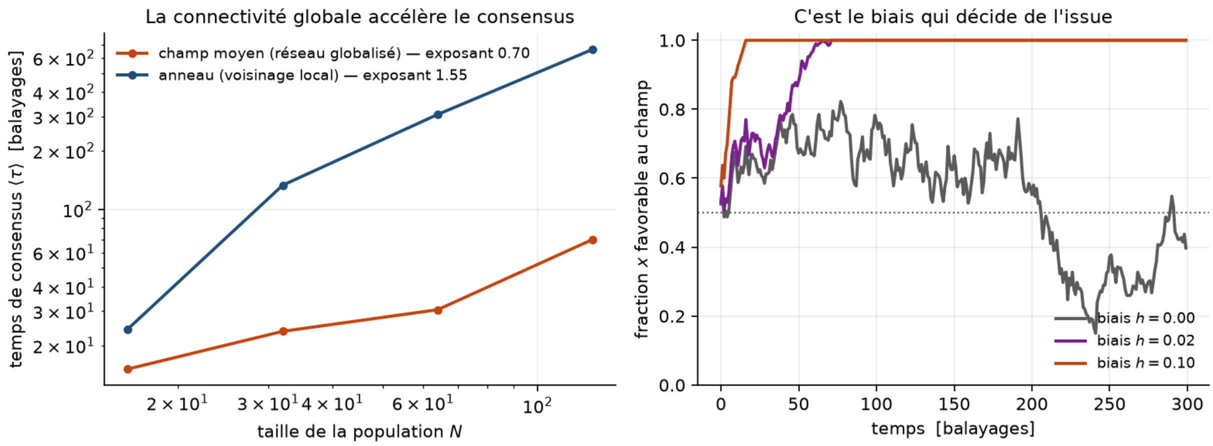

Figure 3: Left: consensus time versus population size, log-log, for two topologies. Right: trajectories of the field-favouring fraction for three bias intensities — it is bias, not connectivity, that decides the outcome.

bias of algorithmic micro-fields and the *homophily* that compartments the graph [Pariser, 2011]. The regulatory consequence is clear: the useful lever acts on the field, not on the number of links.

5 The landscape of public opinion

5.1 The equation and its drift

At macroscopic scale, collective opinion $x \in [-1, 1]$ obeys a Fokker-Planck equation [Risken, 1989]:

$$\frac{\partial P(x,t)}{\partial t} = -\frac{\partial}{\partial x}[A(x)P(x,t)] + \frac{\partial^2}{\partial x^2}[B(x)P(x,t)]. \tag{5}$$

The quadratic potential $V(x) = -\frac{J}{2}x^2 - Hx$, which comes naturally to mind, can produce *no* phase transition: the corresponding exponential is convex, hence maximal at the extremes at any temperature. The missing term is the entropy of mixing — the number of individual configurations compatible with a mean opinion x. Adding it recovers the Helmholtz free energy:

$$f(x) = \underbrace{-\frac{J}{2}x^2 - Hx}_{E} + T\underbrace{\left[\frac{1+x}{2}\ln\frac{1+x}{2} + \frac{1-x}{2}\ln\frac{1-x}{2}\right]}_{-S}, \tag{6}$$

giving the drift

$$A(x) = -f'(x) = Jx + H - T\,\mathrm{artanh}(x). \tag{7}$$

The entropic restoring term bounds the dynamics within $(-1, 1)$ and yields a mean-field critical temperature $T_c = J$. The form $A(x) = Jx + H$ is its linearisation near $x = 0$.

5.2 Diffusion and the effect of N

The diffusion term reads

$$B(x) = \frac{k_B T(1-x^2)}{N}, \tag{8}$$

the factor $(1 - x^2)$ switching off noise at unanimity.

The $1/N$ factor requires careful reading, since it appears to contradict the initial hypothesis: the larger the population, the *less* noisy its macroscopic variable. There is no contradiction, but two distinct scales. Total configurational entropy is extensive and grows as N; fluctuations of the mean x decay as $1/N$. The defensible thesis is therefore:

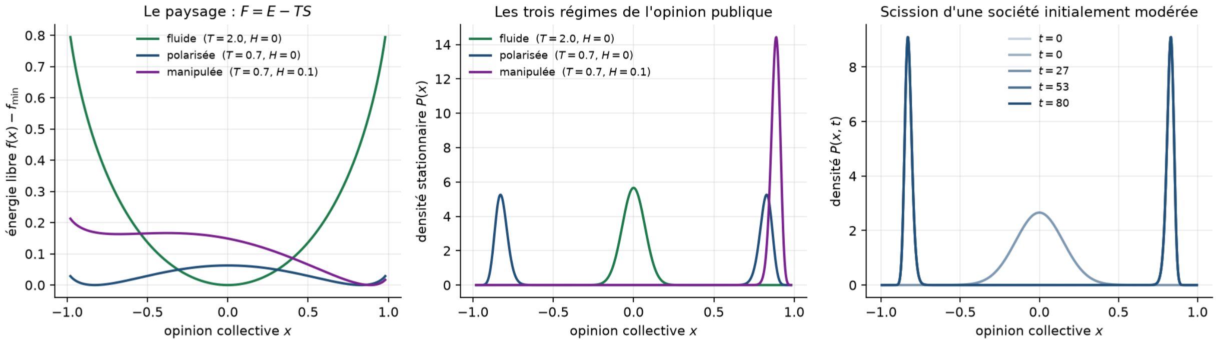

Figure 4: Free-energy landscape (6), the associated stationary distributions, and the temporal splitting of an initially moderate society. The solver uses finite volumes with zero flux at the boundaries: probability mass is conserved to 10^{-15}, which distinguishes a genuine collapse of moderation from numerical leakage.

> A large population does not become noisy, it becomes *rigid*. Its size does not agitate it; it deprives it of stochastic plasticity — and it is that rigidity which makes polarisation irreversible, since a system without noise can no longer leave the potential well it has fallen into.

This reformulation is stronger than the initial hypothesis: it explains irreversibility, which the "entropy pump" metaphor did not.

5.3 Three regimes

The equilibrium distribution takes the large-deviation form $P_{\text{stat}}(x) \propto \exp(-Nf(x)/T)$, where the factor N makes the penalty more severe the larger the population. Three regimes follow, obtained by varying two parameters (figure 4): a fluid society ($T > T_c$), a polarised one ($T < T_c$, $H = 0$) and a manipulated one ($T < T_c$, $H > 0$). In the last case the peak on the field's side overwhelms the other: the "false consensus" is not an added assumption, it is the tilted well.

The density of moderates decays exponentially with N. In a large population below T_c, moderation is not a minority position: it is statistically inaccessible.

5.4 Barrier crossing

Moving from one bloc to the other without passing through moderation is thermally activated barrier crossing, with rate $\propto e^{-\Delta V/k_B T}$ [Kramers, 1940], and not tunnelling — which has no classical counterpart. The distinction is not cosmetic: Kramers' law supplies a temperature dependence, hence a testable prediction.

6 Algorithmic resonance kinetics

A piece of false information and a recommender algorithm form a positive feedback loop analogous to the Larsen effect. Visibility $V(t)$ obeys

$$\ddot{V} + (\lambda - \gamma\alpha\,\sigma(V))\dot{V} + \omega_0^2 V = \xi(t), \tag{9}$$

where λ is natural damping (forgetting), γ the algorithmic gain, α the content's emotional charge, and $\sigma(V) = 1/(1 + (V/V_{\text{sat}})^2)$ a saturation factor expressing the finiteness of available attention.

The instability criterion is

$$\boxed{\gamma\alpha > \lambda} \tag{10}$$

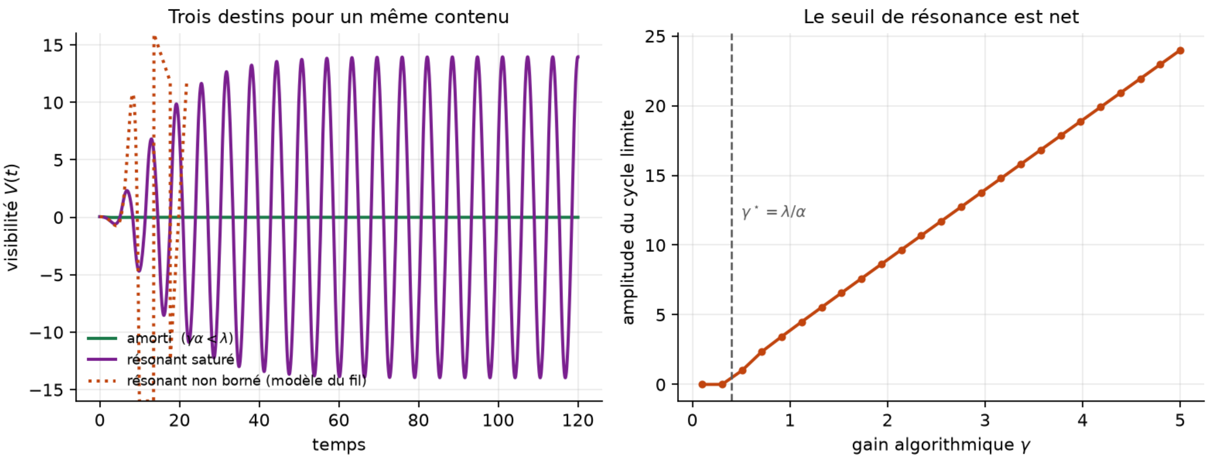

Figure 5: Left: three fates for the same content — damped, bounded resonant, unbounded resonant. Right: limit-cycle amplitude versus algorithmic gain; the measured threshold coincides with $\gamma^\star = \lambda/\alpha$.

Beyond it, effective damping becomes negative: the system accumulates energy instead of dissipating it. Figure 5 confirms that the measured threshold coincides with $\gamma^\star = \lambda/\alpha$.

Without saturation the unstable solution diverges as $e^{(\gamma\alpha-\lambda)t}$, which is exact but unusable: infinite visibility does not exist and no threshold is calibratable on a divergent model. With saturation the system settles into a limit cycle — a self-sustaining media oscillation of finite amplitude, matching the observed behaviour of established disinformation.

One point deserves emphasis for regulation. At *uniform* gain, a more emotional item crosses the threshold where a factual one does not. A platform therefore need not favour disinformation in order to amplify it selectively: the bias is mechanical. This is what makes measuring γ more relevant, and more enforceable, than auditing editorial intent.

7 Two instruments

7.1 The Entropic Dissipation Index

The index is the Shannon entropy of the viewpoints served to a user, normalised by its theoretical maximum:

$$\mathrm{EDI} = \frac{H(X)}{\log_2 k} \in [0, 1]. \tag{11}$$

Normalisation is the essential point: it makes the index dimensionless and comparable across platforms, which a threshold in bits does not. The denominator k — the number of viewpoints the platform is *able* to serve — must be imposed by the regulator, failing which a perfectly closed feed would obtain a flattering index.

A collapsed index is the signature of a zero local social temperature, that is, a frozen state in the Ising sense — the regime in which hysteresis makes false beliefs persistent.

7.2 The Entropic Dissipation Algorithm

A conventional recommender maximises immediate engagement; by (10), that cost function is the one producing negative damping. The algorithm replaces it with

$$S(i, c) = \mathrm{Relevance}(i, c) + \mu \cdot \Delta H(i, c), \qquad \mu \geq 0, \tag{12}$$

where $\Delta H = H_{\text{future}} - H_{\text{current}}$ measures the item's impact on the feed's index. The positive sign is essential: it rewards diversification. The relevance term is retained — an algorithm serving diversity without relevance would be abandoned by its users, dissipating no entropy at all.

The coefficient μ is not constant. It stays at a resting value while the index is satisfactory, and rises progressively once the index falls below the critical threshold: this transposes simulated annealing, a temperature spike to break a frozen state followed by cooling. At $\mu = 0$ the algorithm is indistinguishable from an engagement filter, which makes it a parameterised addition rather than a redesign.

8 Limitations

These are stated here rather than left to reviewers.

No empirical calibration. This is the principal weakness. The parameters J, T, γ, α have no estimation procedure on real data. The formalism is coherent; its empirical grounding remains to be built. The most accessible route would be estimating $\gamma\alpha/\lambda$ by fitting (9) to public visibility time series.

An analogy is not an explanation. Nothing here demonstrates that opinion *obeys* statistical mechanics. The work establishes that a formalism borrowed from physics *reproduces* certain observed behaviours. Kramers' prediction on the temperature dependence of switching rates would offer a falsifiable test; it has not been carried out.

Individuals are not spins. Intentionality, irony and strategic contrarianism have no counterpart in a spin model. Opinions are multidimensional — the agent model uses two continuous axes, which mitigates the problem without solving it [Deffuant et al., 2000]. Finally, the environment is not passive: an algorithm optimises a cost function, making it a strategic actor rather than a thermal bath. This is the analogy's weakest point, and it is also what makes the algorithm conceivable — a cost function can be changed.

Limitations specific to the index as an instrument. Discretisation into viewpoints is a political choice: whoever defines the modalities defines the index. The index is gameable, since a platform can serve formally divergent but substantively empty content. Measuring it on individual feeds raises a privacy question this work does not answer. And imposing an index floor is a constraint on what citizens see: defensible, but an intervention in public debate, and presenting it as a neutral technical measure would be dishonest.

Scope of the simulations. The regimes explored were chosen for legibility. No systematic sensitivity study was carried out, system sizes are modest, and no result is compared against real data. The conclusions are qualitative: they concern the existence of regimes and the direction of dependencies, never transposable numerical values.

9 Reproducibility

The complete work is openly available, runnable in containers, with 203 automated tests. The structuring verifications cover Onsager's critical temperature (4), exact conservation of probability mass, the exponents of consensus-time scaling laws, the strict positivity of the hysteresis cycle area below T_c, and the equivalence of the algorithm with an engagement filter at $\mu = 0$. Every figure in this note is regenerated by a notebook.

https://github.com/s-geffroy/Index-Dissipation-Entropique
https://s-geffroy.github.io/Index-Dissipation-Entropique/en/

This work is open to critical review. Feedback on empirical calibration, on the index's viability as a regulatory instrument, or on remaining analogies that do not hold, is the most valuable.

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